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Microfluidics system, instrument, and cartridge including self-aligning optical fiber system and method

Abstract

The present invention is directed to microfluidics systems, instruments, and cartridges including self-aligning optical fiber systems and methods of use thereof. More specifically, the disclosure describes a microfluidics instrument including an optical detection system, microfluidics cartridge, and a self-aligning optical fiber system capable of coupling the microfluidics instrument and the microfluidics cartridge. Further, the disclosure provides methods of optical detection operations using a microfluidics system.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
5452390	12/1994	Bechtel	385/60	G02B 6/421
6485189	12/2001	Gilliland et al.	N/A	N/A
7553453	12/2008	Gu et al.	N/A	N/A
8940147	12/2014	Bartsch et al.	N/A	N/A
2007/0190525	12/2006	Gu	435/5	G01N 15/1484
2007/0211985	12/2006	Duer	422/82.11	G01N 21/253
2009/0068668	12/2008	Duer	N/A	N/A
2009/0147253	12/2008	Hartmann et al.	N/A	N/A
2022/0341843	12/2021	Anderson	N/A	G01N 21/05

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2814720	12/2011	CA	N/A
20210017067	12/2020	KR	N/A
WO-2006088976	12/2005	WO	B82Y 15/00
WO-2006098700	12/2005	WO	B01L 3/502784
2020049524	12/2019	WO	N/A
2020132146	12/2019	WO	N/A
2021146804	12/2020	WO	N/A
2021146809	12/2020	WO	N/A

OTHER PUBLICATIONS

International Search Report issued in International Patent Application No. PCT/CA2022/050854, mailed Aug. 30, 2022, 9 pages. cited by applicant
Written Opinion of the International Searching Authority issued in International Patent Application No. PCT/CA2022/050854, mailed Aug. 30, 2022, 7 pages. cited by applicant
Hartman et al., A low-cost, manufacturable method for fabricating capillary and optical fiber interconnects for microfluidic devices, Lab on a Chip, Royal Society of Chemistry, Apr. 8, 2008, 8(4):609-616. cited by applicant
European Patent Office, Extended European Search Report, Application No. 22810019.4, Feb. 20, 2025, 10 pages. cited by applicant

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Background/Summary

FIELD OF THE INVENTION
(1) The presently disclosed subject matter relates generally to optical fiber interfaces and more particularly to a microfluidics system, instrument, and cartridge including a self-aligning optical fiber system and method.
RELATED APPLICATIONS
(2) The presently-disclosed subject matter is related to and claims priority to International Application No. PCT/CA2022/050854 entitled “Microfluidics system, instrument, and cartridge including self-aligning optical fiber system and method,” filed on May 26, 2022; U.S. Provisional Patent Application No. U.S. 63/193,944, entitled “Microfluidics Instrument and Cartridge Including an Optical Fiber Alignment Mechanism,” filed on May 27, 2021; and U.S. Provisional Patent Application No. U.S. 63/237,868, entitled “Microfluidics System, Instrument, and Cartridge Including Self-Aligning Optical Fiber System and Method,” filed on Aug. 27, 2021; the entire disclosures of which are herein incorporated by reference.
BACKGROUND
(3) In applications that require optical coupling between two systems, devices, and/or components, it may be difficult to accomplish optical coupling in an easy and automated manner. For example, typical optical fiber couplers are single connectors that are manipulated manually or have screw terminals to make the connection.
(4) In microfluidics applications, for example, optical coupling may be required between a microfluidics instrument and a microfluidics device (or cartridge). In this example, a limitation is that optical fibers may be manually connected to the microfluidics device (or cartridge) one at a time. Further, the number of optical connections may be limited. Another limitation is that the mechanical complexity of optical coupling may lie mostly at the microfluidics device (or cartridge) side of the system and with little or no complexity at the microfluidics instrument side of the system. Therefore, new approaches are needed with respect to optical coupling between two systems, devices, and/or components.
SUMMARY
(5) In some aspects, the present invention is directed to a microfluidics system comprising: (a) a microfluidics instrument, wherein the microfluidics instrument includes an optical detection system; (b) a microfluidics cartridge; and (c) a self-aligning optical fiber system; wherein the self-

aligning optical fiber system optically couples the microfluidics instrument and the microfluidics cartridge. In some embodiments, the optical detection system comprises an illumination source and an optical measurement device.
(6) In some embodiments, the microfluidics system includes a self-aligning optical fiber system comprises an instrument fiber optic coupler and a cartridge fiber optic connector. In some embodiments, the self-aligning optical fiber system comprises a plurality of optical detection channels, wherein each of the plurality of optical detection channels comprises an instrument optical channel and a cartridge optical channel. In one embodiment, the self-aligning optical fiber system comprises from about 4 to about 16 optical detection channels. In one embodiment, each of the instrument optical channels optically connects each of the cartridge optical channels to the optical measurement device.
(7) In some embodiments, the instrument fiber optic coupler comprises a plurality of instrument ferrule assemblies each comprising a leading tip end and an instrument optical fiber. In some embodiments, the cartridge fiber optic connector further comprises a plurality of cartridge ferrule assemblies each comprising a receiving end capable of receiving the instrument ferrule assembly and each comprising a cartridge optical fiber.
(8) In some embodiments, the microfluidics instrument further comprises a movable slide mechanism, and wherein the movable slide mechanism is operable to create an optical coupling between the microfluidics instrument and the microfluidics cartridge by engaging the instrument fiber optic coupler with the cartridge fiber optic connector. In other embodiments, the movable slide mechanism is operable to engage the instrument ferrule assembly to the cartridge ferrule assembly and thereby coupling the instrument optical fibers to the cartridge optical fibers. In still other embodiments, the movable slide mechanism comprises a slidable base plate mounted on a rail, a backplate mounted on the end of slidable base plate, a leadscrew and associated motor operable to advance and/or retract the instrument fiber optic coupler with respect to the cartridge fiber optic connector.
(9) In one embodiment, the microfluidic cartridge further comprises: (a) a bottom substrate, wherein the bottom substrate comprises a droplet operations surface; (b) a top substrate; and wherein the bottom substrate and the top substrate are separated by a droplet operation gap therebetween. In some embodiments, the microfluidics cartridge is a digital microfluidics cartridge (DMF). In some embodiment, the bottom substrate and/or top substrate comprise a PCB substrate, a glass substrate or a silicon substrate, and wherein the PCB substrate, glass substrate, or silicon substrate is optionally coated with a dielectric layer and one or more electrodes operable for droplet operations.
(10) In some embodiments, the droplet operation gap between the bottom substrate and the top substrate is filled with a filler fluid. In one embodiment, the filler fluid is a low-viscosity oil or a halogenated oil.
(11) In some embodiments, the optical detection system comprises one or more surface plasmon resonance (SPR) sensors or one or more localized surface plasmon resonance (LSPR) sensors.
(12) In accordance with another aspect, the present invention is directed to a method for performing an optical detection operation, the method comprising: (a) providing a microfluidics system, wherein the microfluidics system comprises a microfluidics instrument, a microfluidics cartridge and a plurality of optical detection channels, wherein: (i) the microfluidics instrument comprises an instrument fiber optics coupler; and (ii) the microfluidics cartridge comprises a cartridge fiber optics connector; (b) performing a first optical alignment step to align the instrument fiber optic coupler to the cartridge fiber optic connector; (c) performing a second optical alignment step to individually align each optical detection channel; and (d) performing an optical detection operation using the microfluidics system, microfluidics instrument and microfluidics cartridge.
(13) In one embodiment, the microfluidic instrument further comprises a movable slide mechanism, and wherein the first optical alignment step is carried out by moving the movable slide

mechanism until the fiber optics coupler engages the fiber optics connector. In one embodiment, the fiber optics coupler is moved towards a stationary fiber optics connector. In some embodiments, the first optical alignment step results in a course alignment of each of the plurality of optical detection channels.
(14) In one embodiment, the second optical alignment step results in a fine alignment of each of the plurality of optical detection channels. In some embodiments, the second alignment step is carried out by continuing to translate the movable slide mechanism towards the microfluidics cartridge until the fiber optics coupler fully engages the fiber optics connector, and thereby individually aligning each of the optical detection channels.
(15) In some embodiments, the fiber optics coupler further comprises a plurality of instrument ferrule assemblies each comprising a leading tip end and an instrument optical fiber, wherein the fiber optics connector further comprises a plurality of cartridge ferrule assemblies each comprising a receiving end capable of receiving the instrument ferrule assembly and each comprising a cartridge optical fiber, and wherein the movable slide is moved until each of the instrument ferrule assemblies engages each of the cartridge ferrule assemblies, thereby connecting the instrument optical fibers with the cartridge optical fibers and creating the plurality of optical detection channels. In other embodiments, the instrument fiber optic coupler further comprises a housing having one or more dowel pins, wherein the cartridge optic connector further comprises a housing having one or more datum holes and wherein the one or more datum holes accept the one or more dowel pins during the first alignment step. In one embodiment, the dowel pins align instrument fiber optic coupler to cartridge fiber optic connector.
(16) In one embodiment, the second alignment step results in Z-direction alignment between the fiber optics coupler and fiber optics connector. In another embodiment, the second alignment step results in Z-direction alignment between each of the instrument ferrule assemblies and each of the cartridge ferrule assemblies and aligns the instrument optical fibers and the cartridge optical fibers face-to-face substantially without leaving any gap therebetween. In still another embodiment, the instrument ferrule assembly further comprises a spring and wherein the spring aligns the instrument optical fibers and the cartridge optical fibers face-to-face substantially without leaving any gap therebetween.
(17) In one embodiment, the instrument optical fibers are aligned to within about +/−0.7 mm of cartridge optical fibers. In another embodiment, the instrument optical fibers are aligned to within about +/−50 μm of cartridge optical fibers. In some embodiments, an optic gel is applied between the instrument optical fibers and the cartridge optical fibers.
(18) In some embodiments, the microfluidics system further comprises an optical detection system comprising an illumination source and an optical measurement device. In other embodiments, the optical detection system comprises surface plasmon resonance (SPR) or localized surface plasmon resonance (LSPR), and wherein the optical detection system comprises an SPR or LSPR illumination source and one or more SPR or LSPR optical measurement devices. In still other embodiments, wherein the microfluidics cartridge is a digital microfluidics cartridge (DMF).

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) The features and advantages of the present invention will be more clearly understood from the following description taken in conjunction with the accompanying drawings, which are not necessarily drawn to scale, and wherein:
(2) FIG. **1** illustrates a block diagram of a microfluidics system including an example of the presently disclosed self-aligning optical fiber system for coupling optically a microfluidics instrument and a microfluidics device (or cartridge);

(3) FIG. **2**A illustrates a perspective view of an example instantiation of the microfluidics system shown in FIG. **1** and including the microfluidics instrument coupled optically to the microfluidics device (or cartridge) using the presently disclosed self-aligning optical fiber system;
(4) FIG. **2**B illustrates a perspective view of an example of the microfluidics instrument potion of the microfluidics system shown in FIG. **1**;
(5) FIG. **3**, FIG. **4**, and FIG. **5** illustrate perspective views of an example of the presently disclosed self-aligning optical fiber system shown in FIG. **1** and FIG. **2** and including an instrument fiber optic coupler and a cartridge fiber optic connector;
(6) FIG. **6** illustrates a perspective view showing more details of the presently disclosed self-aligning optical fiber system shown in FIG. **1** through FIG. **5**;
(7) FIG. **7**A and FIG. **7**B illustrate a perspective view and an exploded view, respectively, of an example of an instrument ferrule assembly of the instrument fiber optic coupler shown in FIG. **13** and FIG. **14**;
(8) FIG. **8** illustrates various views showing more details of an example of an instrument ferrule of the instrument ferrule assembly shown in FIG. **7**A;
(9) FIG. **9**A and FIG. **9**B illustrate other perspective views of the instrument ferrule assembly shown in FIG. **7**A;
(10) FIG. **10** illustrates a side view of the instrument ferrule assembly shown in FIG. **7**A;
(11) FIG. **11**A and FIG. **11**B illustrate a front view and a back view, respectively, of the instrument ferrule assembly shown in FIG. **7**A;
(12) FIG. **12** illustrates perspective views of the instrument ferrule assembly shown in FIG. **7**A in various states of assembly;
(13) FIG. **13** and FIG. **14** illustrate a perspective view and an exploded view, respectively, of an example of the instrument fiber optic coupler of the presently disclosed self-aligning optical fiber system shown in FIG. **1** through FIG. **6**;
(14) FIG. **15**A illustrates a perspective view of an example of a cartridge ferrule assembly of the cartridge fiber optic connector of the presently disclosed self-aligning optical fiber system shown in FIG. **1** through FIG. **6**;
(15) FIG. **15**B illustrates a perspective cross-sectional view of the cartridge ferrule assembly shown in FIG. **15**A;
(16) FIG. **16**A and FIG. **16**B illustrate a top view and a side view, respectively, of the cartridge ferrule assembly shown in FIG. **15**A;
(17) FIG. **17**A and FIG. **17**B illustrate a front view and a back view, respectively, of the cartridge ferrule assembly shown in FIG. **15**A;
(18) FIG. **18** illustrates an exploded view of the cartridge ferrule assembly shown in FIG. **15**A;
(19) FIG. **19** illustrates a cross-sectional view showing more details of an example of the cartridge ferrule of the cartridge ferrule assembly shown in FIG. **15**A;
(20) FIG. **20** and FIG. **21** illustrate cross-sectional views of a portion of the presently disclosed self-aligning optical fiber system shown in FIG. **1** through FIG. **6** and showing the instrument fiber optic coupler and the cartridge fiber optic connector fully engaged together;
(21) FIG. **22** and FIG. **23** illustrate top views of the instrument fiber optic coupler in relation to the cartridge fiber optic connector of the presently disclosed self-aligning optical fiber system shown in FIG. **1** through FIG. **6**;
(22) FIG. **24** and FIG. **25** illustrate bottom views of the instrument fiber optic coupler in relation to the cartridge fiber optic connector of the presently disclosed self-aligning optical fiber system shown in FIG. **1** through FIG. **6**; and
(23) FIG. **26** illustrates a flow diagram of an example of a method of using of the presently disclosed microfluidics system including the self-aligning optical fiber system shown in FIG. **1** through FIG. **25**.
DEFINITIONS

(24) “Activate,” with reference to one or more electrodes, means affecting a change in the electrical state of the one or more electrodes which, in the presence of a droplet, results in a droplet operation. Activation of an electrode can be accomplished using alternating current (AC) or direct current (DC). Any suitable voltage may be used. For example, an electrode may be activated using a voltage which is greater than about 5 V, or greater than about 20 V, or greater than about 40 V, or greater than about 100 V, or greater than about 200 V or greater than about 300 V. The suitable voltage being a function of the dielectric's properties such as thickness and dielectric constant, liquid properties such as viscosity and many other factors as well. Where an AC signal is used, any suitable frequency may be employed. For example, an electrode may be activated using an AC signal having a frequency from about 1 Hz to about 10 MHz, or from about 1 Hz and 10 KHz, or from about 10 Hz to about 240 Hz, or about 60 Hz.
(25) “Droplet” means a volume of liquid on a droplet actuator. Typically, a droplet is at least partially bounded by a filler fluid. For example, a droplet may be completely surrounded by a filler fluid or may be bounded by filler fluid and one or more surfaces of the droplet actuator. As another example, a droplet may be bounded by filler fluid, one or more surfaces of the droplet actuator, and/or the atmosphere. As yet another example, a droplet may be bounded by filler fluid and the atmosphere. Droplets may, for example, be aqueous or non-aqueous or may be mixtures or emulsions including aqueous and non-aqueous components.
(26) “Droplet Actuator” means a device for manipulating droplets. Microfluidics devices, microfluidics cartridges, digital microfluidics (DMF) devices, and DMF cartridges are examples of droplet actuators. Certain droplet actuators will include one or more substrates arranged with a droplet operations gap therebetween and electrodes associated with (e.g., patterned on, layered on, attached to, and/or embedded in) the one or more substrates and arranged to conduct one or more droplet operations. For example, certain droplet actuators will include a base (or bottom) substrate, droplet operations electrodes associated with the substrate, one or more dielectric layers atop the substrate and/or electrodes, and optionally one or more hydrophobic layers atop the substrate, dielectric layers and/or the electrodes forming a droplet operations surface. A top substrate may also be provided, which is separated from the droplet operations surface by a gap, commonly referred to as a droplet operations gap. Droplet actuators will include various electrode arrangements on the top and/or bottom substrates. During droplet operations it is preferred that droplets remain in continuous contact or frequent contact with a ground or reference electrode. A ground or reference electrode may be associated with the top substrate facing the gap, the bottom substrate facing the gap, or within the gap itself. Where electrodes are provided on both substrates, electrical contacts for coupling the electrodes to a droplet actuator instrument for controlling or monitoring the electrodes may be associated with one or both plates. In some cases, electrodes on one substrate are electrically coupled to the other substrate so that only one substrate is in contact with the droplet actuator. Where multiple substrates are used, a spacer may be provided between the substrates to determine the height of the gap therebetween and define on-actuator dispensing reservoirs. The spacer height may, for example, be from about 5 μm to about 1000 μm, or about 100 μm to about 400 μm, or about 200 μm to about 350 μm, or about 250 μm to about 300 μm, or about 275 μm. The spacer may, for example, be formed of features or layers projecting from the top or bottom substrates, and/or a material inserted between the top and bottom substrates. One or more openings may be provided in the one or more substrates for forming a fluid path through which liquid may be delivered into the droplet operations gap.
(27) In some cases, the top and/or bottom substrate of a droplet actuator includes a PCB substrate that is coated with a dielectric, such as a polyimide dielectric, which may in some cases also be coated or otherwise treated to make the droplet operations surface hydrophobic. Various materials are also suitable for use as the dielectric component of the droplet actuator. In some cases, the top and/or bottom substrate of a droplet actuator includes a glass or silicon substrate on which features have been patterned using process technology borrowed from semiconductor device fabrication

including the deposition and etching of thin layers of materials using microlithography. The top and/or bottom substrate may consist of a semiconductor backplane (i.e., a thin-film transistor (TFT) active-matrix controller) on which droplet operations electrodes have been formed.

(28) Electrodes of a droplet actuator are typically controlled by a controller or a processor, which is itself provided as part of a system, which may include processing functions as well as data and software storage and input and output capabilities. Reagents may be provided on the droplet actuator in the droplet operations gap or in a reservoir fluidly coupled to the droplet operations gap. The reagents may be in liquid form, e.g., droplets, or they may be provided in a reconstitutable form in the droplet operations gap or in a reservoir fluidly coupled to the droplet operations gap. Reconstitutable reagents may typically be combined with liquids for reconstitution.

(29) “Droplet operation” means any manipulation of a droplet on a droplet actuator. A droplet operation may, for example, include: loading a droplet into the droplet actuator; dispensing one or more droplets from a source droplet; splitting, separating or dividing a droplet into two or more droplets; transporting a droplet from one location to another in any direction; merging or combining two or more droplets into a single droplet; diluting a droplet; mixing a droplet; agitating a droplet; deforming a droplet; retaining a droplet in position; incubating a droplet; heating a droplet; vaporizing a droplet; cooling a droplet; disposing of a droplet; transporting a droplet out of a droplet actuator; other droplet operations described herein; and/or any combination of the foregoing. The terms “merge,” “merging,” “combine,” “combining” and the like are used to describe the creation of one droplet from two or more droplets. It should be understood that when such a term is used in reference to two or more droplets, any combination of droplet operations that are sufficient to result in the combination of the two or more droplets into one droplet may be used. For example, “merging droplet A with droplet B,” can be achieved by transporting droplet A into contact with a stationary droplet B, transporting droplet B into contact with a stationary droplet A, or transporting droplets A and B into contact with each other. The terms “splitting,” “separating” and “dividing” are not intended to imply any particular outcome with respect to volume of the resulting droplets (i.e., the volume of the resulting droplets can be the same or different) or number of resulting droplets (the number of resulting droplets may be 2, 3, 4, 5 or more). The term “mixing” refers to droplet operations which result in more homogenous distribution of one or more components within a droplet. Examples of “loading” droplet operations include microdialysis loading, pressure assisted loading, robotic loading, passive loading, and pipette loading. Droplet operations may be electrode-mediated. In some cases, droplet operations are further facilitated by the use of hydrophilic and/or hydrophobic regions on surfaces and/or by physical obstacles. For examples of droplet operations, see the patents and patent applications cited above under the definition of “droplet actuator.” Impedance and/or capacitance sensing and/or imaging techniques may sometimes be used to determine or confirm the outcome of a droplet operation. Generally speaking, the sensing or imaging techniques may be used to confirm the presence or absence of a droplet at a specific electrode. For example, the presence of a dispensed droplet at the destination electrode following a droplet dispensing operation confirms that the droplet dispensing operation was effective. Similarly, the presence of a droplet at a detection spot at an appropriate step in an assay protocol may confirm that a previous set of droplet operations has successfully produced a droplet for detection. Droplet transport time can be quite fast. For example, in various embodiments, transport of a droplet from one electrode to the next may be completed within about 1 sec, or about 0.1 sec, or about 0.01 sec, or about 0.001 sec. In one embodiment, the electrode is operated in AC mode but is switched to DC mode for imaging. It is helpful for conducting droplet operations for the footprint area of droplet to be similar to or larger than the electrowetting area; in other words, 1×-, 2×-3×-droplets are usefully controlled and/or operated using 1, 2, and 3 electrodes, respectively. If the droplet footprint is greater than number of electrodes available for conducting a droplet operation at a given time, the difference between the droplet size and the number of electrodes should typically not be greater than 1; in other words, a 2× droplet is usefully

controlled using 1 electrode and a 3× droplet is usefully controlled using 2 electrodes. When droplets include beads, it is useful for droplet size to be equal to the number of electrodes controlling the droplet, e.g., transporting the droplet.
(30) “Filler fluid” means a fluid associated with a droplet operations substrate of a droplet actuator, which fluid is sufficiently immiscible with a droplet phase to render the droplet phase subject to electrode-mediated droplet operations. For example, the droplet operations gap of a droplet actuator is typically filled with a filler fluid. The filler fluid may, for example, be or include a low-viscosity oil, such as silicone oil or hexadecane. The filler fluid may be or include a halogenated oil, such as a fluorinated or perfluorinated oil. The filler fluid may fill the entire gap of the droplet actuator or may only coat one or more surfaces of the droplet actuator. Filler fluids may be selected to improve droplet operations and/or reduce loss of reagent or target substances from droplets, reduce formation of unwanted microdroplets, reduce cross contamination between droplets, reduce contamination of droplet actuator surfaces, reduce degradation of droplet actuator materials, reduce evaporation of droplets, etc. For example, filler fluids may be selected for compatibility with droplet actuator materials. As an example, fluorinated filler fluids may be usefully employed with fluorinated surface coatings. Fluorinated filler fluids are useful to reduce loss of lipophilic compounds, such as umbelliferone substrates like 6-hexadecanoylamido-4-methylumbelliferone substrates (e.g., for use in Krabbe, Niemann-Pick, or other assays); Filler fluids may, for example, be doped with surfactants or other additives. For example, additives may be selected to improve droplet operations and/or reduce loss of reagent or target substances from droplets, formation of microdroplets, cross contamination between droplets, contamination of droplet actuator surfaces, degradation of droplet actuator materials, etc. Composition of the filler fluid, including surfactant doping, may be selected for performance with reagents or samples used in the specific assay protocols and effective interaction or non-interaction with droplet actuator materials. For example, fluorinated oils may in some cases be doped with fluorinated surfactants, e.g., Zonyl FSO-100 (Sigma-Aldrich) and/or others.
(31) The terms “top,” “bottom,” “over,” “under,” and “on” are used throughout the description with reference to the relative positions of components of the droplet actuator, such as relative positions of top and bottom substrates of the droplet actuator. It will be appreciated that in many cases the droplet actuator is functional regardless of its orientation in space.
(32) When a liquid in any form (e.g., a droplet or a continuous body, whether moving or stationary) is described as being “on”, “at”, or “over” an electrode, array, matrix or surface, such liquid could be either in direct contact with the electrode/array/matrix/surface, or could be in contact with one or more layers or films that are interposed between the liquid and the electrode/array/matrix/surface. In one example, filler fluid can be considered as a dynamic film between such liquid and the electrode/array/matrix/surface.
(33) When a droplet is described as being “on” or “loaded on” a droplet actuator, it should be understood that the droplet is arranged on the droplet actuator in a manner which facilitates using the droplet actuator to conduct one or more droplet operations on the droplet, the droplet is arranged on the droplet actuator in a manner which facilitates sensing of a property of or a signal from the droplet, and/or the droplet has been subjected to a droplet operation on the droplet actuator.

DETAILED DESCRIPTION OF THE INVENTION

(34) The presently disclosed subject matter now will be described more fully hereinafter with reference to the accompanying drawings, in which some, but not all embodiments of the presently disclosed subject matter are shown. Like numbers refer to like elements throughout. The presently disclosed subject matter may be embodied in many different forms and should not be construed as limited to the embodiments set forth herein; rather, these embodiments are provided so that this disclosure will satisfy applicable legal requirements. Indeed, many modifications and other embodiments of the presently disclosed subject matter set forth herein will come to mind to one

skilled in the art to which the presently disclosed subject matter pertains having the benefit of the teachings presented in the foregoing descriptions and the associated drawings. Therefore, it is to be understood that the presently disclosed subject matter is not to be limited to the specific embodiments disclosed and that modifications and other embodiments are intended to be included within the scope of the appended claims.

(35) In some embodiments, the presently disclosed subject matter provides a microfluidics system, instrument, and cartridge including a self-aligning optical fiber system and method.

(36) In some embodiments, the presently disclosed microfluidics system, instrument, and cartridge including a self-aligning optical fiber system and method provide a mechanism that enables the interface between a microfluidics instrument and a disposable microfluidics device (or cartridge) that includes embedded optical fiber sensors.

(37) In some embodiments, the presently disclosed microfluidics system, instrument, and cartridge including a self-aligning optical fiber system and method provide an instrument fiber optic coupler on the microfluidics instrument side of the microfluidics system and a cartridge fiber optic connector on the microfluidics device (or cartridge) side of the microfluidics system.

(38) In some embodiments, the presently disclosed microfluidics system, instrument, and cartridge including a self-aligning optical fiber system and method provide an instrument fiber optic coupler on the microfluidics instrument side of the microfluidics system and a cartridge fiber optic connector on the microfluidics device (or cartridge) side of the microfluidics system that ensure a good optical mate (about <10 um spacing, about <100 um concentricity deviation) therebetween.

(39) In some embodiments, the presently disclosed microfluidics system, instrument, and cartridge including a self-aligning optical fiber system and method provide an instrument fiber optic coupler on the microfluidics instrument side of the microfluidics system and a cartridge fiber optic connector on the microfluidics device (or cartridge) side of the microfluidics system that ensure a good optical mate therebetween and across a significant distance allowing easy to manufacture tolerances on the disposable microfluidics device to enable mass-manufacturing.

(40) In some embodiments, the presently disclosed microfluidics system, instrument, and cartridge including a self-aligning optical fiber system and method provide an instrument fiber optic coupler on the microfluidics instrument side of the microfluidics system and a cartridge fiber optic connector on the microfluidics device (or cartridge) side of the microfluidics system and wherein the self-aligning optical fiber system may support any number of optical detection channels, such as, but not limited to, sixteen (16) optical detection channels. In general, any number of optical detection channels can be used. For example, in some embodiments, the self-aligning optical fiber system may comprise from about four (4) to about sixteen (16), from about four (4) to about fourteen (14), or from about six (6) to about twelve (12) optical detection channels. In other embodiments, the self-aligning optical fiber system may comprise 4, 6, 8, 10, 12, 14, or 16 optical detection channels.

(41) In some embodiments, the presently disclosed microfluidics system, instrument, and cartridge including a self-aligning optical fiber system and method provide an instrument fiber optic coupler on the microfluidics instrument side of the microfluidics system and a cartridge fiber optic connector on the microfluidics device (or cartridge) side of the microfluidics system that engage and align in one or two stages: (1) a course alignment stage that aligns the instrument fiber optic coupler to the cartridge fiber optic connector and/or (2) a fine alignment stage that aligns individually each optical channel of the instrument fiber optic coupler and the cartridge fiber optic connector.

(42) In some embodiments, the presently disclosed microfluidics system, instrument, and cartridge including a self-aligning optical fiber system and method provide an instrument fiber optic coupler on the microfluidics instrument side of the microfluidics system that may include a line or arrangement of multiple (e.g., sixteen) instrument ferrule assemblies and wherein each of the instrument ferrule assemblies may include an off-the-shelf ferrule. In general, any number of

instrument ferrule assemblies can be used. For example, in some embodiments, the microfluidics system may comprise a line or arrangement of from about four (4) to about sixteen (16), from about four (4) to about fourteen (14), from about (6) to about twelve (12) instrument ferrule assemblies. In other embodiments, the microfluidics system may comprise 4, 6, 8, 10, 12, 14, or 16 instrument ferrule assemblies. Furthermore, in some embodiments, each of the instrument ferrule assemblies includes an instrument optical fiber.

(43) In some embodiments, the presently disclosed microfluidics system, instrument, and cartridge including a self-aligning optical fiber system and method provide a cartridge fiber optic connector on the microfluidics device (or cartridge) side of the microfluidics system that may include a line or arrangement of multiple (e.g., sixteen) cartridge ferrule assemblies and wherein each of the cartridge ferrule assemblies may include a cup-shaped custom ferrule that is designed to accept the off-the-shelf ferrule of the instrument ferrule assemblies and implements the optical fiber fine alignment. In general, any number of cartridge ferrule assemblies can be used. For example, in some embodiments, the microfluidics system may comprise a line or arrangement of from about four (4) to about sixteen (16), from about four (4) to about fourteen (14), from about (6) to about twelve (12) cartridge ferrule assemblies. In still other embodiments, the microfluidics system may comprise 4, 6, 8, 10, 12, 14, or 16 cartridge ferrule assemblies. Furthermore, in some embodiments, each of the cartridge ferrule assemblies includes a cartridge optical fiber.

(44) In some embodiments, the presently disclosed microfluidics system, instrument, and cartridge including a self-aligning optical fiber system and method provide an instrument fiber optic coupler on the microfluidics instrument side of the microfluidics system and a cartridge fiber optic connector on the microfluidics device (or cartridge) side of the microfluidics system for aligning a series of optical fibers (e.g., sixteen) in the cartridge fiber optic connector simultaneously onto the same number of optical fibers (e.g., sixteen) in the instrument fiber optic coupler to transmit the optic result concurrently through the established optical channels for diagnostics in the microfluidics system and/or instrument.

(45) In some embodiments, the presently disclosed microfluidics system, instrument, and cartridge and method may provide a self-aligning optical fiber system in which the tolerance for aligning a line of multiple optical fibers across some distance lies substantially entirely in each individual mating of one instrument ferrule assembly to one cartridge ferrule assembly, not in the collective arrangement of, for example, sixteen instrument ferrule assemblies mating to sixteen cartridge ferrule assemblies across some distance (although other arrangements are contemplated herein). While the presently disclosed self-aligning optical fiber system may be described herein with reference to a microfluidics system, instrument, and cartridge, the presently disclosed self-aligning optical fiber system may not be limited to microfluidics applications only. This is exemplary only. The presently disclosed self-aligning optical fiber system may be used in any applications requiring optical coupling and/or interfaces between two systems, devices, and/or components.

(46) Additionally, while the presently disclosed self-aligning optical fiber system may be described herein with reference to supporting sixteen (16) optical detection channels in a microfluidics system, instrument, and cartridge, the presently disclosed self-aligning optical fiber system may not be limited to supporting sixteen (16) optical detection channels only. This is exemplary only. The presently disclosed self-aligning optical fiber system may be provided to support any number of optical detection channels, as described elsewhere herein.

(47) Referring now to FIG. **1** is a block diagram of a microfluidics system **100** including an example of the presently disclosed self-aligning optical fiber system for coupling optically a microfluidics instrument and a microfluidics device (or cartridge). In this example, microfluidics system **100** may include a self-aligning optical fiber system **110**. Self-aligning optical fiber system **110** further includes an instrument fiber optic coupler **112** on the instrument side of microfluidics system **100** and a cartridge fiber optic connector **114** on the microfluidics cartridge side of microfluidics system **100**.

(48) For example, microfluidics system **100** may include a microfluidics instrument **160** and a microfluidics cartridge **170** that may be coupled optically using self-aligning optical fiber system **110**. In this example, instrument fiber optic coupler **112** of self-aligning optical fiber system **110** may be provided at microfluidics instrument **160**. Further, cartridge fiber optic connector **114** of self-aligning optical fiber system **110** may be provided at microfluidics cartridge **170**.
(49) Microfluidics instrument **160** may further include an optical detection system **162** and a movable slide mechanism **164**. Instrument fiber optic coupler **112** at microfluidics instrument **160** may include an arrangement of optical fibers **154**, such as sixteen optical fibers **154**. The sixteen optical fibers **154** may run from instrument fiber optic coupler **112** to optical detection system **162** via a fiber optic bundle **166**.
(50) Optical detection system **162** of microfluidics instrument **160** may be, for example, an optical measurement system that includes an illumination source (not shown) and an optical measurement device (not shown). For example, optical detection system **162** may be a fluorimeter that provides both excitation and detection. In this example, the illumination source (e.g., a light source for the visible range (400-800 nm)) and the optical measurement device (e.g., charge coupled device, photodetector, spectrometer, photodiode array) may be arranged with respect to microfluidics cartridge **170**. Further, microfluidics system **100** is not limited to one optical detection system **162** only (e.g., one illumination source and one optical measurement device only). Microfluidics system **100** may include multiple optical detection systems **162** (e.g., multiple illumination sources and/or multiple optical measurement devices) to support multiple detection spots **178** of microfluidics cartridge **170**.
(51) Movable slide mechanism **164** of microfluidics instrument **160** may be any mechanism for sliding microfluidics instrument **160** toward the stationary microfluidics cartridge **170** that are arranged in the same plane. The sliding action of movable slide mechanism **164** is used to engage instrument fiber optic coupler **112** at microfluidics instrument **160** with cartridge fiber optic connector **114** at microfluidics cartridge **170**. In this way, optical coupling occurs between microfluidics instrument **160** and microfluidics cartridge **170**. That is, optical pathways or channels are provided from detection spots **178** of microfluidics cartridge **170** to optical detection system **162** of microfluidics instrument **160**. An example of movable slide mechanism **164** is shown in FIG. **2**A.
(52) Microfluidics cartridge **170** may be, for example, any disposable or non-disposable digital microfluidics (DMF) device (or cartridge), droplet actuator device (or cartridge), droplet operations device (or cartridge), and the like. Cartridge fiber optic connector **114** may include an arrangement of optical fibers **146**, such as sixteen optical fibers **146**. The sixteen optical fibers **146** may run from cartridge fiber optic connector **114** to sixteen respective sensors **176** and/or sixteen respective detection spots **178** of microfluidics cartridge **170**.
(53) In one example, sensors **176** may be surface plasmon resonance (SPR) sensors that support the detection spots **178** (i.e.; detection channels) of microfluidics cartridge **170**. In this example, each SPR sensor **176** may be a functionalized SPR sensor **176** (i.e., ligands immobilized on the surface).
(54) In another example, sensors **176** may be localized surface plasmon resonance (LSPR) sensors. In this example, each LSPR sensor **176** may be a functionalized for (1) detecting, for example, certain molecules (e.g., target analytes) and/or chemicals in the sample, and (2) analysis of analytes; namely, for measuring binding events in real time to extract ON-rate information, OFF-rate information, and/or affinity information.
(55) In one example, detection spots **178** of microfluidics cartridge **170** may be certain droplet operations electrodes (i.e., electrowetting electrodes, not shown) dedicated to optical detection operations of microfluidics system **100**. More details of sensors **176** and detection spots **178** of microfluidics cartridge **170** are shown and described hereinbelow with reference to FIG. **6**.
(56) In self-aligning optical fiber system **110**, optical fibers **154** of instrument fiber optic coupler **112** align with and couple optically to the respective optical fibers **146** of cartridge fiber optic

connector **114**. More details of the optical coupling mechanisms of self-aligning optical fiber system **110** are shown and described hereinbelow with reference to FIG. **2**A through FIG. **26**.

(57) Further, self-aligning optical fiber system **110** may not be limited to supporting sixteen (16) optical detection channels only, such as the sixteen optical fibers **154** of instrument fiber optic coupler **112** coupled optically to the sixteen optical fibers **146** of cartridge fiber optic connector **114**. Self-aligning optical fiber system **110** including instrument fiber optic coupler **112** and cartridge fiber optic connector **114** may be designed to support any number of optical detection channels.

(58) Referring now to FIG. **2**A is a perspective view of an example instantiation of microfluidics system **100** shown in FIG. **1** and including microfluidics instrument **160** coupled optically to microfluidics cartridge **170** using the presently disclosed self-aligning optical fiber system **110**.

(59) In this example, movable slide mechanism **164** may include a slidable base plate **210** mounted on a rail **212**, a backplate **214** mounted on the end of slidable base plate **210** furthest from microfluidics cartridge **170**. Additionally, movable slide mechanism **164** may include a motor **216** that is held stationary with respect to slidable base plate **210** and backplate **214**. A leadscrew **218** of motor **216** may be threaded through a threaded hole of backplate **214**. Generally, motor **216** may be used to advance and/or retract instrument fiber optic coupler **112** at microfluidics instrument **160** with respect to cartridge fiber optic connector **114** at microfluidics cartridge **170**.

(60) Additionally, instrument fiber optic coupler **112** of self-aligning optical fiber system **110** is arranged at the end of slidable base plate **210** that is opposite backplate **214**. Instrument fiber optic coupler **112** may be held to the end of slidable base plate **210** using a screw or fastener **220** and spring **222** at each side of slidable base plate **210** to engage each end of instrument fiber optic coupler **112**. In this arrangement, screws or fasteners **220** float on springs **222**, screws or fasteners **220** are not used to pull and engage instrument fiber optic coupler **112** to movable slide mechanism **164**. Rather, screws or fasteners **220** provide sliding guides while the screw action of motor **216** and leadscrew **218** may be used to translate slidable base plate **210** in the same plane as microfluidics cartridge **170**. In doing so, the totality of movable slide mechanism **164**, instrument fiber optic coupler **112**, and cartridge fiber optic connector **114** of microfluidics instrument **160** are pulled together and engaged.

(61) FIG. **2**A also shows that instrument fiber optic coupler **112** may include a coupler housing **116** that has a dowel pin **118** at each side. In one example, dowel pins **118** may be 1.6 mm diameter dowel pins. Further, cartridge fiber optic connector **114** may include connector housing **132** that has a cartridge datum hole **134**. The two cartridge datum holes **134** at connector housing **132** of cartridge fiber optic connector **114** are designed to receive the two dowel pins **118** at coupler housing **116** of instrument fiber optic coupler **112**. Further to the example, FIG. **2**B shows a perspective view of an example of the microfluidics instrument potion of microfluidics system **100** shown in FIG. **1**. For example, FIG. **2**B shows more details of a cable **211** for providing fiber optic bundle **166** to instrument fiber optic coupler **112**.

(62) More details of align instrument fiber optic coupler **112** and cartridge fiber optic connector **114** are shown and described hereinbelow with reference to FIG. **3** through FIG. **25**.

(63) In microfluidics system **100**, instrument fiber optic coupler **112** of microfluidics instrument **160** and cartridge fiber optic connector **114** of microfluidics instrument **160** may engage and align in two stages: (1) a course alignment stage that aligns instrument fiber optic coupler **112** to cartridge fiber optic connector **114** and (2) a fine alignment stage that aligns individually each optical channel of instrument fiber optic coupler **112** and cartridge fiber optic connector **114**. The process of using movable slide mechanism **164** may be used to perform the first of the two stages, which is the course alignment stage that uses dowel pins **118** to align instrument fiber optic coupler **112** to cartridge fiber optic connector **114**. In this course alignment stage, optical fibers **154** (not visible) of instrument fiber optic coupler **112** may be aligned (end-to-end or face-to-face) to within about +/−0.7 mm of optical fibers **146** (not visible) of cartridge fiber optic connector **114**.

(64) Referring now to FIG. **3**, FIG. **4**, and FIG. **5** is perspective views of an example of the presently disclosed self-aligning optical fiber system **110** shown in FIG. **1** and FIG. **2**A and including instrument fiber optic coupler **112** and cartridge fiber optic connector **114**. Further, FIG. **3**, FIG. **4**, and FIG. **5** show instrument fiber optic coupler **112** in relation to cartridge fiber optic connector **114**, but not yet mechanically and/or optically coupled. Additionally, FIG. **3**, FIG. **4**, and FIG. **5** show more clearly the two dowel pins **118** at coupler housing **116** of instrument fiber optic coupler **112** in relation to the two cartridge datum holes **134** at connector housing **132** of cartridge fiber optic connector **114**. Additionally, FIG. **3** and FIG. **4** show fastener holes **122** at the ends of coupler housing **116** (e.g., one on each end of coupler housing **116**) for receiving screws or fasteners **220** of slidable base plate **210**.
(65) In this example, coupler housing **116** of instrument fiber optic coupler **112** may hold an arrangement of instrument ferrule assemblies **124** that include optical fibers **154** (not visible). In this example, coupler housing **116** holds sixteen instrument ferrule assemblies **124** for holding optical fibers **154**, arranged in a line. Similarly, connector housing **132** of cartridge fiber optic connector **114** may hold an arrangement of cartridge ferrule assemblies **136** that include optical fibers **146** (not visible). In this example, connector housing **132** holds sixteen cartridge ferrule assemblies **136** for holding optical fibers **146**, arranged in a line.
(66) Additionally, microfluidics cartridge **170** may include a bottom substrate **172** and a top substrate **174** separated by a droplet operations gap (not shown). In one example, bottom substrate **172** may b a printed circuit board (PCB) and top substrate **174** may be a substantially transparent glass or plastic substrate. Further, connector housing **132** of cartridge fiber optic connector **114** may be integrated into top substrate **174** of microfluidics cartridge **170**. For example, connector housing **132** and top substrate **174** may be a one-piece molded plastic design. Further, bottom substrate **172** may include any arrangements of droplet operations electrodes (i.e., electrowetting electrodes, not shown) for conducting droplet operations on a droplet operations surface.
(67) Referring now to FIG. **6** is a perspective view showing more details of the presently disclosed self-aligning optical fiber system **110** shown in FIG. **1** through FIG. **5**. For example, FIG. **6** shows only the arrangement of instrument ferrule assemblies **124** of instrument fiber optic coupler **112**. More details of instrument fiber optic coupler **112** and instrument ferrule assembly **124** are shown and described hereinbelow with reference to FIG. **7**A through FIG. **14**.
(68) Further, FIG. **6** shows microfluidics cartridge **170** absent top substrate **174** and showing more details of cartridge ferrule assemblies **136**. For example, each cartridge ferrule assembly **136** may include a cartridge ferrule **138** holding an optical fiber **146**. A distal end **150** of optical fiber **146** may include a sensor **176** (e.g., SPR sensor **176** and/or LSPR sensor **176**). Further, distal end **150** of optical fiber **146** with sensor **176** may be positioned at a certain detection spot **178**, which is a certain detection channel of microfluidics cartridge **170**. In another example, the distal end **150** of optical fiber **146** may be absent sensor **176** and, instead, sensor **176** may be provided at detection spot **178** as a part of microfluidics cartridge **170** and/or droplet operations. More details of cartridge ferrule assembly **136** are shown and described hereinbelow with reference to FIG. **15**A through FIG. **19**.
(69) Referring now to FIG. **7**A and FIG. **7**B is a perspective view and an exploded view, respectively, of an example of instrument ferrule assembly **124** of instrument fiber optic coupler **112** shown in FIG. **13** and FIG. **14**. In this example, each instrument ferrule assembly **124** of instrument fiber optic coupler **112** may include an instrument ferrule **152** that has a leading tip **153**, an optical fiber **154** encased in cable cladding **155**, a flat washer **156**, and a spring **158**.
(70) In one example, instrument ferrule **152** may be a commercially available ferrule, such as a ferrule available from Precision Fiber Products, Inc (PFP) (Chula Vista, CA), in the family of Part Number: MM-FER2007CF-XXXX. FIG. **8** show various views of an example of the PFP instrument ferrule **152** that may be used in instrument ferrule assembly **124**. In this example, instrument ferrule **152** may have an overall length of about 11.80 mm. The outside diameter (OD)

of leading tip **153** may be about 1.25 mm. The end face of leading tip **153** may be conical shaped. The bore size of instrument ferrule **152** for receiving optical fiber **154** may be, for example, from about 80 μm to about 126 μm. Further, in instrument ferrule **152**, the ferrule material may be, for example, ceramic zirconia (ZrO2) and the flange material may be, for example, nickel plated brass.
(71) In instrument ferrule assembly **124**, the OD of optical fiber **154** may be, for example, from about 100 μm to about 125 μm. The OD of cable cladding **155** that surrounds optical fiber **154** may be, for example, about 0.125+0/−0.008 mm.
(72) In instrument ferrule assembly **124**, flat washer **156** may be an M2 polyslider washer that is, for example, from about 0.4 mm to about 0.5 mm thick with an OD of about 3.5 mm.
(73) In instrument ferrule assembly **124**, spring **158** may be a coiled wire type spring with ground ends.
(74) In instrument ferrule assembly **124** and referring still to FIG. **7**A and FIG. **7**B, flat washer **156** may be provided against the back of the flange portion of instrument ferrule **152**. Then, spring **158** may be provided against the back of flat washer **156**. Then, an exposed portion of optical fiber **154** may be inserted into instrument ferrule **152** and advanced to leading tip **153** of instrument ferrule **152**. Then, some length of cable cladding **155** engages inside the rear portion of instrument ferrule **152**. When instrument ferrule assembly **124** is fully assembled, an end face of optical fiber **154** may be exposed at an opening in the front face of leading tip **153** of instrument ferrule **152**. Accordingly, optical fiber **154** of instrument ferrule assembly **124** may be coupled optically to a corresponding optical fiber **146** of a corresponding cartridge ferrule assembly **136**. An example of which is shown hereinbelow with reference to FIG. **20** and FIG. **21**.
(75) Referring now to FIG. **9**A and FIG. **9**B is other perspective views of instrument ferrule assembly **124** shown in FIG. **7**A. Further, FIG. **10** shows a side view of instrument ferrule assembly **124**. Further, FIG. **11**A and FIG. **11**B shows a front view and a back view, respectively, of instrument ferrule assembly **124**. FIG. **9**A through FIG. **11**B show that instrument ferrule assembly **124** may further include a shoulder washer **128** against the leading face of the flange portion of instrument ferrule **152**. That is, shoulder washer **128** may be provided on leading tip **153** of instrument ferrule **152**. Shoulder washer **128** may be, for example, an M2 shoulder countersunk washer.
(76) Referring now to FIG. **12** is perspective views of instrument ferrule assembly **124** shown in FIG. **7**A in various states of assembly. For example, instrument ferrule assembly **124** is shown absent spring **158**. Flat washer **156** is shown alone in relation to optical fiber **154**. Shoulder washer **128** is shown alone in relation to optical fiber **154**. Instrument ferrule **152** is shown alone in relation to optical fiber **154**. Spring **158** is shown alone in relation to optical fiber **154**. Optical fiber **154** is shown alone. Then, instrument ferrule assembly **124** is shown fully assembled.
(77) Referring now to FIG. **13** and FIG. **14** is a perspective view and an exploded view, respectively, of an example of instrument fiber optic coupler **112** of the presently disclosed self-aligning optical fiber system **110** shown in FIG. **1** through FIG. **6**. FIG. **13** and FIG. **14** show that instrument fiber optic coupler **112** may further include a coupler clamp plate **120** for holding the multiple instrument ferrule assemblies **124** secure to coupler housing **116**. Further, coupler clamp plate **120** may be secured to coupler housing **116** via multiple screws **126**.
(78) Referring now to FIG. **15**A is a perspective view of an example of cartridge ferrule assembly **136** of cartridge fiber optic connector **114** of the presently disclosed self-aligning optical fiber system **110** shown in FIG. **1** through FIG. **6**. In this example, cartridge ferrule assembly **136** may include cartridge ferrule **138** and optical fiber **146**. Cartridge ferrule **138** may further include a receiving end **140** and a fiber holder end **142**. Receiving end **140** of cartridge ferrule **138** may be cup-shaped and sized for receiving leading tip **153** of instrument ferrule **152** of instrument ferrule assembly **124**.
(79) In cartridge ferrule assembly **136**, the OD of optical fiber **146** may be, for example, from about 200 μm to about 230 μm. The OD of the cable cladding (not shown) that surrounds optical fiber

146 may be, for example, about 0.23+0/−0.008 mm.
(80) Optical fiber **146** may have a proximal end **148** and distal end **150**. Proximal end **148** of optical fiber **146** may be fitted into a fiber centering channel **144** at receiving end **140** of cartridge ferrule **138**, as shown in FIG. **15**B, which is a cross-sectional view of cartridge ferrule assembly **136** taken along line A-A of FIG. **15**A. A length of optical fiber **146** runs through fiber holder end **142** of cartridge ferrule **138** and with distal end **150** of optical fiber **146** extending outward away from fiber holder end **142**.
(81) Additionally, FIG. **16**A and FIG. **16**B shows a top view and a side view, respectively, of cartridge ferrule assembly **136**. Further, FIG. **17**A and FIG. **17**B shows a front view and a back view, respectively, of cartridge ferrule assembly **136**. Further, FIG. **18** shows an exploded view of cartridge ferrule assembly **136**.
(82) Referring now to FIG. **19** is a cross-sectional view showing more details of an example of cartridge ferrule **138** of cartridge ferrule assembly **136** shown in FIG. **15**A. In this example, the overall length of cartridge ferrule **138** may be about 7.4 mm. The overall diameter of receiving end **140** of cartridge ferrule **138** may be about 3.1 mm. The ID of receiving end **140** of cartridge ferrule **138** may be about 1.25 mm. The overall cross-section of fiber holder end **142** of cartridge ferrule **138** may be about 1.6 mm. The ID of fiber holder end **142** of cartridge ferrule **138** may be about 2 mm. The ID of fiber centering channel **144** leading into receiving end **140** of cartridge ferrule **138** may be about 0.23 mm.
(83) Additionally, a lead-in bevel or chamfer may be provided at the openings of both receiving end **140** and fiber holder end **142** of cartridge ferrule **138**. The lead-in bevel or chamfer may be used to provide a generous lead-in of, for example, from about 0.8 mm to about 0.9 mm during the mating between leading tip **153** of instrument ferrule **152** and receiving end **140** of cartridge ferrule **138** during alignment.
(84) Fiber centering channel **144** is designed to hold proximal end **148** of optical fiber **146**. When instrument ferrule assembly **124** is fully assembled, an end face of optical fiber **146** may be exposed at an opening in the floor of receiving end **140** of cartridge ferrule **138**. Accordingly, optical fiber **146** of cartridge ferrule assembly **136** may be coupled optically to a corresponding optical fiber **154** of a corresponding instrument ferrule assembly **124**. That is, an optical fiber **146** of a cartridge ferrule assembly **136** and an optical fiber **154** of an instrument ferrule assembly **124** may come face-to-face or end-to-end inside the cup-shaped receiving end **140** of cartridge ferrule **138**. An example of which is shown hereinbelow with reference to FIG. **20** and FIG. **21**.
(85) Referring now to FIG. **20** and FIG. **21** is cross-sectional views of a portion of the presently disclosed self-aligning optical fiber system **110** shown in FIG. **1** through FIG. **6** and showing instrument fiber optic coupler **112** and cartridge fiber optic connector **114** fully engaged together.
(86) In self-aligning optical fiber system **110**, the OD of optical fiber **146** of cartridge ferrule assembly **136** may be larger than the OD of optical fiber **154** of instrument ferrule assembly **124**. For example, the OD of optical fiber **146** may be from about 200 μm to about 230 μm, while the OD of optical fiber **154** may be from about 100 μm to about 125 μm. This allows a certain tolerance for the end of optical fiber **154** of instrument ferrule assembly **124** aligning with and matching up with the end of optical fiber **146** of cartridge ferrule assembly **136**. In self-aligning optical fiber system **110**, this configuration with respect to optical fibers **146** and **154** provides an alignment tolerance of better than about +/−50 μm.
(87) Again, in microfluidics system **100**, instrument fiber optic coupler **112** of microfluidics instrument **160** and cartridge fiber optic connector **114** of microfluidics instrument **160** may engage and align in two stages: (1) a course alignment stage that aligns instrument fiber optic coupler **112** to cartridge fiber optic connector **114** and (2) a fine alignment stage that aligns individually each optical channel of instrument fiber optic coupler **112** and cartridge fiber optic connector **114**. As previously described, dowel pins **118** of align instrument fiber optic coupler **112** may be used to perform the coarse alignment stage of instrument fiber optic coupler **112** to cartridge fiber optic

connector **114**, which is the first of the two stages.

(88) As shown in FIG. **20** and FIG. **21**, the process of engaging leading tip **153**, which is carrying optical fiber **154**, of instrument ferrule **152** of instrument ferrule assembly **124** with receiving end **140** of cartridge ferrule **138** of cartridge ferrule assembly **136** may be used to perform the second of the two stages, which is the fine alignment stage of instrument fiber optic coupler **112** to cartridge fiber optic connector **114**. In this fine alignment stage, optical fibers **154** of instrument fiber optic coupler **112** may be aligned (end-to-end or face-to-face) to within about +/−50 μm of optical fibers **146** of cartridge fiber optic connector **114**.

(89) FIG. **20** and FIG. **21** show a Z-direction alignment and coupling region **180**. Further, the alignment of optical fiber **154** of instrument fiber optic coupler **112** to optical fiber **146** of cartridge fiber optic connector **114** in Z-direction alignment and coupling region **180** forms an optical channel **182** of microfluidics system **100**. Further, the spring load compression of spring **158** of instrument ferrule assembly **124** of instrument fiber optic coupler **112** allows instrument ferrule **152** and cartridge ferrule **138** to overlap in the z-direction and allows the fiber ends to mate face-to-face substantially without leaving any gap therebetween.

(90) Additionally, while cartridge ferrule **138** may be clamped down using a plate with screw mount, instrument ferrule **152** may be allowed to float by leaving a gap **184** between coupler clamp plate **120** and instrument ferrule **152**. Coupler clamp plate **120** allows installation of instrument ferrule **152**, prevents instrument ferrule **152** from falling out during transportation, and limits the play in the float. The floating instrument ferrule **152** creates the compliance system that allows individual self-aligning ferrules between microfluidics cartridge **170** and microfluidics instrument **160**.

(91) In some embodiments, an optic gel may be applied over the ferrules interface between the ends of the mating two optical fibers **146** and **154**. This, coupled with the z-direction overlap with the spring load compression of spring **158**, may be used to ensure the quality of the optic transmission. A recessed ring (not shown) around the mating surface of cartridge ferrule **138** and a vent hole may allow the optic gel and trapped air to escape during the mating operation.

(92) Referring now to FIG. **22** and FIG. **23** is top views of instrument fiber optic coupler **112** in relation to cartridge fiber optic connector **114** of the presently disclosed self-aligning optical fiber system **110** shown in FIG. **1** through FIG. **6**. Further, FIG. **24** and FIG. **25** is bottom views of instrument fiber optic coupler **112** in relation to cartridge fiber optic connector **114** of the presently disclosed self-aligning optical fiber system **110**.

(93) Referring now to FIG. **26** is a flow diagram of an example of a method **300** of using of the presently disclosed microfluidics system **100** including self-aligning optical fiber system **110** shown in FIG. **1** through FIG. **25**. Method **300** may include, but is not limited to, the following steps.

(94) At a step **310**, the microfluidics system, microfluidics instrument, and/or microfluidics cartridge including the presently disclosed self-aligning optical fiber system is provided. For example, microfluidics system **100** may be provided that includes microfluidics instrument **160**, microfluidics cartridge **170**, and including the presently disclosed self-aligning optical fiber system **110**, as shown and described hereinabove with reference to FIG. **1** through FIG. **25**.

(95) At a step **315**, an instrument fiber optic coupler at the microfluidics instrument is provided in relation to the cartridge fiber optic connector at the microfluidics cartridge. For example, in self-aligning optical fiber system **110**, instrument fiber optic coupler **112** at microfluidics instrument **160** is provided in relation to cartridge fiber optic connector **114** at microfluidics cartridge **170**.

(96) At a step **320**, the first, or coarse optical alignment step of the instrument fiber optic coupler at the microfluidics instrument to the cartridge fiber optic connector at the microfluidics cartridge is performed. For example, the coarse optical alignment step of instrument fiber optic coupler **112** at microfluidics instrument **160** to cartridge fiber optic connector **114** at microfluidics cartridge **170** is performed. For example, movable slide mechanism **164** of microfluidics instrument **160** to may be

used to translate instrument fiber optic coupler **112** toward a stationary cartridge fiber optic connector **114** at microfluidics cartridge **170**. In so doing, the course alignment step uses dowel pins **118** for the initial engagement and alignment of instrument fiber optic coupler **112** to cartridge fiber optic connector **114**. In this course alignment step, optical fibers **154** of instrument fiber optic coupler **112** may be aligned (end-to-end or face-to-face) to within about +/−0.7 mm of optical fibers **146** of cartridge fiber optic connector **114**.

(97) At a step **325**, the second, or fine optical alignment step of the instrument fiber optic coupler at the microfluidics instrument to the cartridge fiber optic connector at the microfluidics cartridge is performed. For example, movable slide mechanism **164** of microfluidics instrument **160** to may be used to continue translating instrument fiber optic coupler **112** toward a stationary cartridge fiber optic connector **114** at microfluidics cartridge **170**. In so doing, the fine optical alignment step may be accomplished by fully engaging instrument ferrule **152** (with optical fibers **154**) of instrument ferrule assembly **124** with cartridge ferrule **138** (with optical fibers **146**) of cartridge ferrule assembly **136**. In this fine alignment step, optical fibers **154** of instrument fiber optic coupler **112** may be aligned (end-to-end or face-to-face) to within about +/−50 μm of optical fibers **146** of cartridge fiber optic connector **114**.

(98) At a step **330**, the optical detection operations of the microfluidics system, microfluidics instrument, and/or microfluidics cartridge using presently disclosed self-aligning optical fiber system are performed. For example, the optical detection operations of microfluidics system **100**, microfluidics instrument **160**, and/or microfluidics cartridge **170** using presently disclosed self-aligning optical fiber system **110** may be performed. For example, microfluidics cartridge **170** may be used to perform DMF operations (or droplet operations) with respect to processing of biological materials. In these processes, optical detection operations may occur wherein self-aligning optical fiber system **110** provides the optical pathways or channels from detection spots **178** of microfluidics cartridge **170** to optical detection system **162** of microfluidics instrument **160**.

(99) Referring now again to FIG. **1** through FIG. **26**, the presently disclosed microfluidics system **100**, microfluidics instrument **160**, microfluidics cartridge **170**, and method **300** may provide self-aligning optical fiber system **110** that may include instrument fiber optic coupler **112** on the microfluidics instrument side of microfluidics system **100** and cartridge fiber optic connector **114** on the microfluidics cartridge side of microfluidics system **100**.

(100) Referring now again to FIG. **1** through FIG. **26**, the presently disclosed microfluidics system **100**, microfluidics instrument **160**, microfluidics cartridge **170**, and method **300** may provide self-aligning optical fiber system **110** in which instrument fiber optic coupler **112** on the microfluidics instrument side of microfluidics system **100** and cartridge fiber optic connector **114** on the microfluidics cartridge side of microfluidics system **100** ensure a good optical mate (about <10 um spacing, about <100 um concentricity deviation) therebetween.

(101) Referring now again to FIG. **1** through FIG. **26**, the presently disclosed microfluidics system **100**, microfluidics instrument **160**, microfluidics cartridge **170**, and method **300** may provide self-aligning optical fiber system **110** in which instrument fiber optic coupler **112** on the microfluidics instrument side of microfluidics system **100** and cartridge fiber optic connector **114** on the microfluidics cartridge side of microfluidics system **100** may support any number of optical detection channels, such as, but not limited to, sixteen (16) optical detection channels.

(102) Referring now again to FIG. **1** through FIG. **26**, the presently disclosed microfluidics system **100**, microfluidics instrument **160**, microfluidics cartridge **170**, and method **300** may provide self-aligning optical fiber system **110** in which instrument fiber optic coupler **112** on the microfluidics instrument side of microfluidics system **100** and cartridge fiber optic connector **114** on the microfluidics cartridge side of microfluidics system **100** may engage and align in two stages: (1) a course alignment stage that aligns instrument fiber optic coupler **112** to cartridge fiber optic connector **114** and (2) a fine alignment stage that aligns individually each optical channel of instrument fiber optic coupler **112** and cartridge fiber optic connector **114**.

(103) Referring now again to FIG. **1** through FIG. **26**, the presently disclosed microfluidics system **100**, microfluidics instrument **160**, microfluidics cartridge **170**, and method **300** may provide self-aligning optical fiber system **110** in which instrument fiber optic coupler **112** on the microfluidics instrument side of microfluidics system **100** and cartridge fiber optic connector **114** on the microfluidics cartridge side of microfluidics system **100** may include a line or arrangement of multiple (e.g., sixteen) instrument ferrule assemblies **124** and wherein each of the instrument ferrule assemblies **124** may include an off-the-shelf ferrule.
(104) Referring now again to FIG. **1** through FIG. **26**, the presently disclosed microfluidics system **100**, microfluidics instrument **160**, microfluidics cartridge **170**, and method **300** may provide self-aligning optical fiber system **110** in which instrument fiber optic coupler **112** on the microfluidics instrument side of microfluidics system **100** and cartridge fiber optic connector **114** on the microfluidics cartridge side of microfluidics system **100** may include a line or arrangement of multiple (e.g., sixteen) cartridge ferrule assemblies **136** and wherein each of the cartridge ferrule assemblies **136** may include a cup-shaped custom cartridge ferrule **138** that is designed to accept the off-the-shelf ferrule of the instrument ferrule assemblies **124** and implements the optical fiber fine alignment.
(105) Referring now again to FIG. **1** through FIG. **26**, the presently disclosed microfluidics system **100**, microfluidics instrument **160**, microfluidics cartridge **170**, and method **300** may provide self-aligning optical fiber system **110** in which the tolerance for aligning a line of multiple optical fibers across some distance lies substantially entirely in each individual mating of one instrument ferrule assembly **124** to one cartridge ferrule assembly **136**, not in the collective arrangement of, for example, sixteen instrument ferrule assemblies **124** mating to sixteen cartridge ferrule assemblies **136** across some distance.
(106) Referring now again to FIG. **1** through FIG. **26**, the presently disclosed microfluidics system **100**, microfluidics instrument **160**, microfluidics cartridge **170**, and method **300** may provide self-aligning optical fiber system **110** in which instrument fiber optic coupler **112** on the microfluidics instrument side of microfluidics system **100** and cartridge fiber optic connector **114** on the microfluidics cartridge side of microfluidics system **100** may be used for aligning a series of optical fibers (e.g., sixteen) in the cartridge fiber optic connector **114** simultaneously onto the same number of optical fibers (e.g., sixteen) in the instrument ferrule assemblies **124** to transmit the optic result concurrently through the established optical channels for diagnostics in the microfluidics system **100** and/or microfluidics instrument **160**.
(107) Referring now again to FIG. **1** through FIG. **26**, microfluidics system **100** and method **300** including the presently disclosed self-aligning optical fiber system **110** may provide a simple and cost-effective way of automating a multiple fibers alignment operation in microfluidics cartridge **170** and microfluidics instrument **160** to establish various channels for optic transmission. This innovation places only a single low-cost injection molded polymer part on the disposable cartridge side and keeps all mechanics on the instrument-side.
(108) Referring now again to FIG. **1** through FIG. **26**, microfluidics system **100** and method **300** including the presently disclosed self-aligning optical fiber system **110** may provide a mechanism wherein each individual cartridge ferrule assembly **136** of microfluidics cartridge **170** may be processed independently prior to cartridge integration, without affecting the mechanism alignment. This is possible because each optical fiber channel may be individually aligned and, accordingly, part-part variation may not affect it. Independent processing is advantageous because it enables novel chemistries to be applied to each individual sensor **176** of its corresponding cartridge ferrule assembly **136**.
(109) Following long-standing patent law convention, the terms “a,” “an,” and “the” refer to “one or more” when used in this application, including the claims. Thus, for example, reference to “a subject” includes a plurality of subjects, unless the context clearly is to the contrary (e.g., a plurality of subjects), and so forth.

(110) The terms “comprise,” “comprises,” “comprising,” “include,” “includes,” and “including,” are intended to be non-limiting, such that recitation of items in a list is not to the exclusion of other like items that may be substituted or added to the listed items.
(111) Terms like “preferably,” “commonly,” and “typically” are not utilized herein to limit the scope of the claimed embodiments or to imply that certain features are critical or essential to the structure or function of the claimed embodiments. These terms are intended to highlight alternative or additional features that may or may not be utilized in a particular embodiment of the present disclosure.
(112) The term “substantially” is utilized herein to represent the inherent degree of uncertainty that may be attributed to any quantitative comparison, value, measurement, or other representation and to represent the degree by which a quantitative representation may vary from a stated reference without resulting in a change in the basic function of the subject matter at issue.
(113) Various modifications and variations of the disclosed methods, compositions and uses of the invention will be apparent to the skilled person without departing from the scope and spirit of the invention. Although the invention has been disclosed in connection with specific preferred aspects or embodiments, it should be understood that the invention as claimed should not be unduly limited to such specific aspects or embodiments.
(114) For the purposes of this specification and appended claims, unless otherwise indicated, all numbers expressing amounts, sizes, dimensions, proportions, shapes, formulations, parameters, percentages, quantities, characteristics, and other numerical values used in the specification and claims, are to be understood as being modified in all instances by the term “about” even though the term “about” may not expressly appear with the value, amount or range. Accordingly, unless indicated to the contrary, the numerical parameters set forth in the following specification and attached claims are not and need not be exact, but may be approximate and/or larger or smaller as desired, reflecting tolerances, conversion factors, rounding off, measurement error and the like, and other factors known to those of skill in the art depending on the desired properties sought to be obtained by the presently disclosed subject matter. For example, the term “about,” when referring to a value can be meant to encompass variations of, in some embodiments ±100%, in some embodiments ±50%, in some embodiments ±20%, in some embodiments ±10%, in some embodiments ±5%, in some embodiments ±1%, in some embodiments ±0.5%, and in some embodiments ±0.1% from the specified amount, as such variations are appropriate to perform the disclosed methods or employ the disclosed compositions.
(115) Further, the term “about” when used in connection with one or more numbers or numerical ranges, should be understood to refer to all such numbers, including all numbers in a range and modifies that range by extending the boundaries above and below the numerical values set forth. The recitation of numerical ranges by endpoints includes all numbers, e.g., whole integers, including fractions thereof, subsumed within that range (for example, the recitation of 1 to 5 includes 1, 2, 3, 4, and 5, as well as fractions thereof, e.g., 1.5, 2.25, 3.75, 4.1, and the like) and any range within that range.
(116) Although the foregoing subject matter has been described in some detail by way of illustration and example for purposes of clarity of understanding, it will be understood by those skilled in the art that certain changes and modifications can be practiced within the scope of the appended claims.
(117) Although the foregoing subject matter has been described in some detail by way of illustration and example for purposes of clarity of understanding, it will be understood by those skilled in the art that certain changes and modifications can be practiced within the scope of the appended claims.

Claims

1. A microfluidics system comprising: (a) a microfluidics instrument, wherein the microfluidics instrument includes an optical detection system comprising a plurality of instrument optical fiber segments; (b) a microfluidics cartridge comprising a plurality of cartridge optical fiber segments; and (c) a self-aligning optical fiber system; wherein the self-aligning optical fiber system is configured to optically couple the plurality of instrument optical fiber segments of the microfluidics instrument and the plurality of cartridge optical fiber segments of the microfluidics cartridge.
2. The microfluidics system of claim 1, wherein the self-aligning optical fiber system comprises an instrument fiber optic coupler and a cartridge fiber optic connector.
3. The microfluidics system of claim 2, wherein the optical detection system comprises an illumination source and an optical measurement device.
4. The microfluidics system of claim 3, wherein self-aligning optical fiber system comprises a plurality of optical detection channels, and wherein each of the plurality of optical detection channels comprises an instrument optical channel and a cartridge optical channel.
5. The microfluidics system of claim 4, wherein each of the instrument optical channels optically connects each of the cartridge optical channels to the optical measurement device.
6. The microfluidics system of claim 5, wherein the self-aligning optical fiber system comprises from about 4 to about 16 optical detection channels.
7. The microfluidics system of claim 6, wherein the instrument fiber optic coupler comprises a plurality of instrument ferrule assemblies each comprising a leading tip end and an instrument optical fiber segment of the plurality of instrument optical fiber segments.
8. The microfluidics system of claim 7, wherein the cartridge fiber optic connector further comprises a plurality of cartridge ferrule assemblies each comprising a receiving end capable of receiving an instrument ferrule assembly of the plurality of instrument ferrule assemblies and each comprising a cartridge optical fiber segment of the plurality of cartridge optical fiber segments.
9. The microfluidics system of claim 8, wherein the microfluidics instrument further comprises a movable slide mechanism, and wherein the movable slide mechanism is operable to create an optical coupling between the microfluidics instrument and the microfluidics cartridge by engaging the instrument fiber optic coupler with the cartridge fiber optic connector.
10. The microfluidics system of claim 9, wherein the movable slide mechanism is operable to engage the plurality of instrument ferrule assemblies to the plurality of cartridge ferrule assemblies and thereby coupling the plurality of instrument optical fiber segments to the plurality of cartridge optical fiber segments.
11. The microfluidics system of claim 10, wherein the movable slide mechanism comprises a slidable base plate mounted on a rail, a backplate mounted on the end of slidable base plate, a leadscrew and associated motor operable to advance and/or retract the instrument fiber optic coupler with respect to the cartridge fiber optic connector.
12. The microfluidics system of claim 1, wherein the microfluidic cartridge further comprises: (a) a bottom substrate, wherein the bottom substrate comprises a droplet operations surface; (b) a top substrate; and wherein the bottom substrate and the top substrate are separated by a droplet operation gap therebetween.
13. The microfluidics system of claim 12, wherein the bottom substrate or top substrate comprise a PCB substrate, a glass substrate or a silicon substrate, and wherein the PCB substrate, glass substrate, or silicon substrate is optionally coated with a dielectric layer and one or more electrodes operable for droplet operations.
14. The microfluidics system of claim 13, wherein the droplet operation gap between the bottom substrate and the top substrate is filled with a filler fluid.
15. The microfluidics system of claim 14, wherein the filler fluid is a low-viscosity oil or a halogenated oil.

16. The microfluidics system of claim 15, wherein the microfluidics cartridge is a digital microfluidics cartridge (DMF).
17. The microfluidics system of claim 16, wherein the optical detection system comprises one or more surface plasmon resonance (SPR) sensors or one or more localized surface plasmon resonance (LSPR) sensors.
18. The microfluidics system of claim 1, wherein the plurality of instrument optical fiber segments are aligned to within about 0.7 mm of the plurality of cartridge optical fiber segments.
19. The microfluidics system of claim 1, wherein the plurality of instrument optical fiber segments are aligned to within about 0.05 mm of the plurality of cartridge optical fiber segments.
